

# Yash Kherwal

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careerautomated.in

## EDUCATION

### Indian Institute of Technology Roorkee

2022 – 2026

*B.Tech in Chemical Engineering*

## EXPERIENCE

### AI Engineer Intern

Feb 2026 – Apr 2026

*Orange Labs — EdTech ERP platform for schools and colleges*

- Built **CCTV-based automated attendance pipeline** — computer vision model detects and identifies students from live camera feeds, auto-syncing presence records into the school ERP; eliminated manual marking.
- Engineered **AI-generated personalised lecture pipeline**: synthesised **teacher-cloned video lectures** from real teacher audio/video, improving engagement over generic video content.

### Software Development Intern

May 2025 – June 2025

*ScoreMe Solutions*

- Designed a **two-stage ML classification pipeline** for fintech PDF processing — rule-based heuristics handle deterministic inputs first; **Random Forest classifier** resolves ambiguous cases, automating **~80% of document volume** and avoiding unnecessary model inference on clear-cut documents.
- Implemented **confidence calibration layer**: predictions below threshold withheld from **credit scoring pipelines** and flagged for review; integrated **Tesseract OCR** as fallback for scanned inputs.

### Engineering Intern

June 2025 – July 2025

*Bharat Electronics Limited*

- Implemented a **binary stream parser** for **ASTERIX CAT048** radar protocol per EUROCONTROL spec; designed **concurrent ingestion** with **backpressure handling** to maintain throughput under sustained high-frequency surveillance loads.

## PROJECTS

### CareerAutomated — Autonomous AI Recruiting Platform | *Python, Playwright, LangGraph, RAG*

2025 – 2026

- Built an end-to-end AI job platform: **50+ ATS discovery connectors** ingest and normalise **1.3M+ live job postings**; a **hybrid BM25 + semantic RAG** pipeline scores per-candidate job fit against real profile data.
- Shipped an **LLM-based resume tailoring engine** under an **integrity gate** that fact-checks every rewritten claim against the source resume, blocking hallucinated content before it ships.
- Engineered a **plugin-based auto-apply engine** — a shared Playwright orchestration core with per-ATS handlers (Greenhouse/Lever/Ashby) fills and submits real applications; **human-in-the-loop safety gates** block submission rather than guess when required data is missing.

### Hybrid RAG Retrieval System | *Python, LangChain, BGE-M3, AstraDB, FastAPI, AWS EC2*

2024

- Diagnosed keyword retrieval failure on 500+ document corpus — BM25 missed semantically relevant results due to vocabulary mismatch; introduced **hybrid retrieval** combining **BGE-M3** dense embeddings with sparse BM25 scores, improving recall on paraphrased and domain-specific queries.
- Designed **decoupled ingestion and serving** on **AWS EC2** for zero-downtime corpus updates; **response groundedness** enforced via prompt constraints to prevent hallucination.

### YAAR — Personality-Adaptive LLM System | *React Native, LLM APIs, FastAPI, Persistent Memory*

2025

- Designed a **persistent personality memory system**: each interaction updates a **personality metric vector** (humour tolerance, self-awareness index, reaction intensity) injected into the **LLM context window** at inference time to steer tone without fine-tuning.
- Shipped production **Android APK** via React Native + EAS Build — validated end-to-end on device with sub-2s streamed LLM response latency.

## TECHNICAL SKILLS

**LLM & Agentic AI:** LangGraph, LangChain, CrewAI, Multi-Agent Orchestration, MCP, Structured Outputs, Prompt Engineering, Agentic Workflows

**Retrieval:** Hybrid RAG, BGE-M3, Dense + Sparse Retrieval, AstraDB, Vector Databases, ANN Indexing

**ML / AI:** TensorFlow, Scikit-learn, Random Forest, Confidence Calibration, Computer Vision (OpenCV)

**Backend & Infra:** FastAPI, Docker, AWS EC2, n8n, REST APIs, Stream Processing

**Languages & Data:** Python, SQL, Java, C++, DSA, Pandas, NumPy